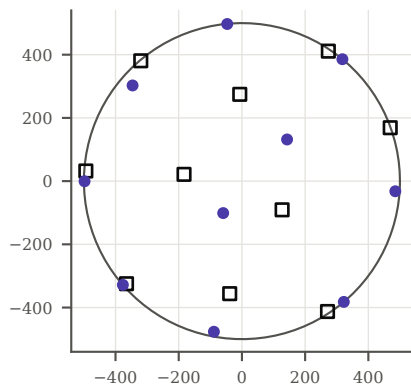


□ best layout of all methods      ● best PSO-VNS layout

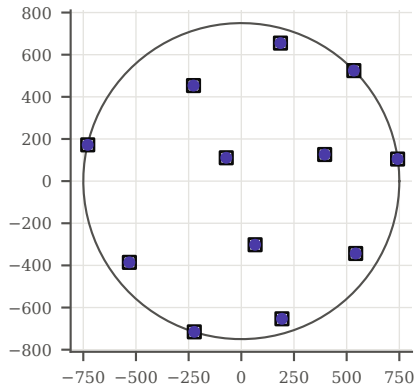
Data Set I,  $r = 500$  m,  $N = 10$

best of all methods: MS-SLSQP (best PSO-VNS: +0.21 pp loss)



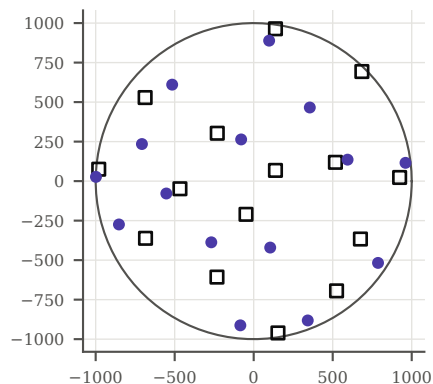
Data Set I,  $r = 750$  m,  $N = 12$

best of all methods: PSO-VNS



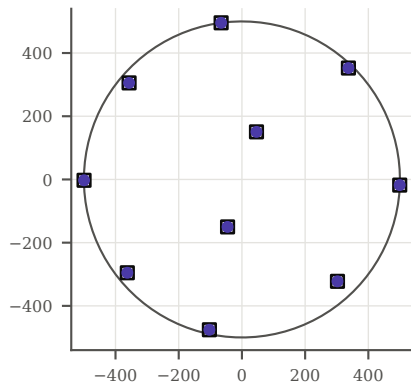
Data Set I,  $r = 1000$  m,  $N = 15$

best of all methods: PSO (best PSO-VNS: +0.09 pp loss)



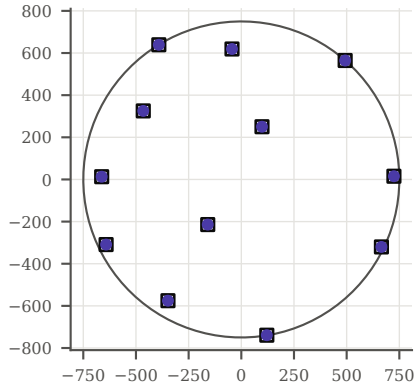
Data Set II,  $r = 500$  m,  $N = 10$

best of all methods: PSO-VNS



Data Set II,  $r = 750$  m,  $N = 12$

best of all methods: PSO-VNS



Data Set II,  $r = 1000$  m,  $N = 15$

best of all methods: PSO-VNS

